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Unit 2 - Understanding Sensors and Lights

PDF Documents

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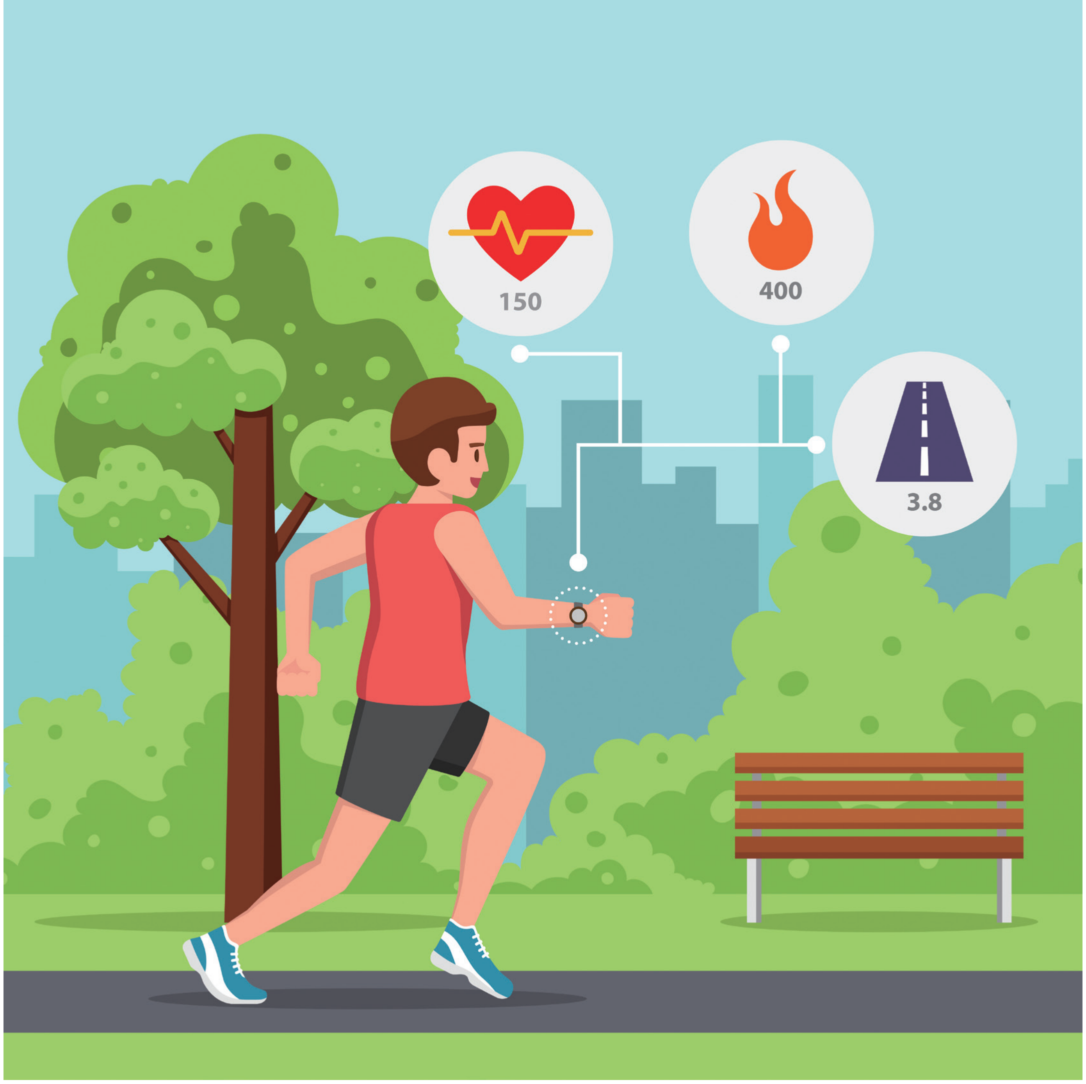
UNIT 2 UNDERSTANDING SENSORS AND LIGHTS

HOW SENSORS WORK

Sensors are like special tools that help us learn about the world around us. They can detect things and send information to other parts of a system. Imagine that you are in parking lot and the sensor tells you a empty spot. When sensor sense the car, it change the light green to red as occipied signal.

Key Points:

In the picture, the watch can calculate heartbeat of the man and distances that he ran. Likewise, we can put it on the sensors to our body.



DETECTION Sensors can sense different things like motion, light, or temperature. For example, a motion sensor can tell if someone is moving close to it.

SIGNALS Once a sensor detects something, it sends a signal. This signal is a message that tells other parts of the system what to do.

RESPONSE The system uses the signal to perform an action. For instance, if a motion sensor detects movement, it might turn on a light.

THE ROLE OF LIGHTS IN INDICATORS

Lights are important parts of indicators. Indicators are devices that show us if something is on or off. Lights help us understand what is happening without having to check everything manually.

Key Points:

VISIBILITY Lights make it easy to see the status of something. For example, if you have a red light on a restroom indicator, it shows that the restroom is occupied.

SIGNALS Lights can signal different messages. For instance, a green light might mean "go" or "safe," while a red light means "stop" or "occupied."

USAGE In a restroom indicator, lights show if the restroom is free or in use. This helps people know whether they can enter or need to wait.

Activity: Make your own Sensor

Let's imagine that you are a scientist who makes the sensor. What kind of sensor do you want to make? Draw about it and write how it works.